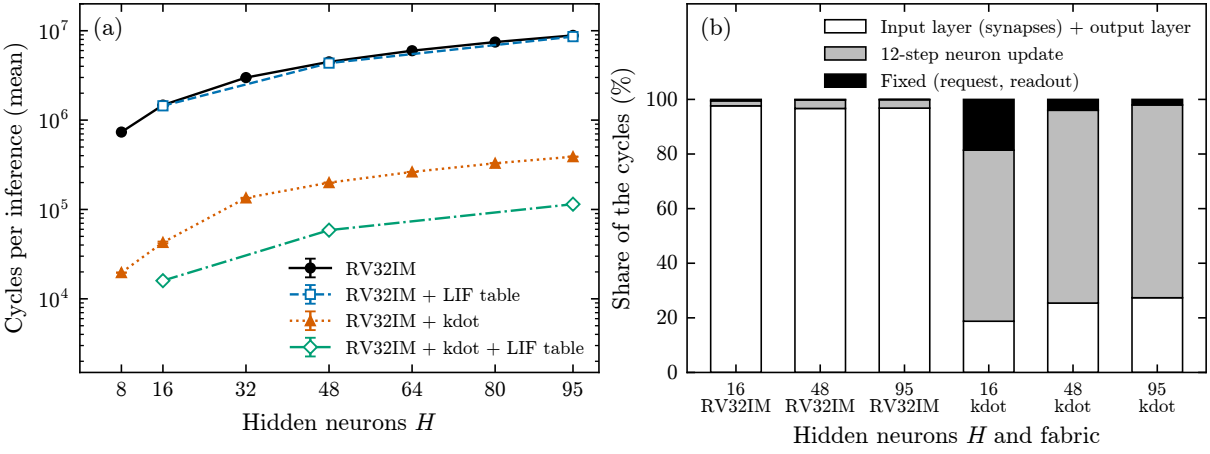
(a)
Cycles per inference (mean)
10^7
10^6
10^5
10^4
RV32IM
RV32IM + LIF table
RV32IM + kdot
RV32IM + kdot + LIF table
8
16
32
48
64
80
95
Hidden neurons H
(b)
Share of the cycles (%)
Input layer (synapses) + output layer
12-step neuron update
Fixed (request, readout)
0
20
40
60
80
100
16 RV32IM
48 RV32IM
95 RV32IM
16 kdot
48 kdot
95 kdot
Hidden neurons H and fabric